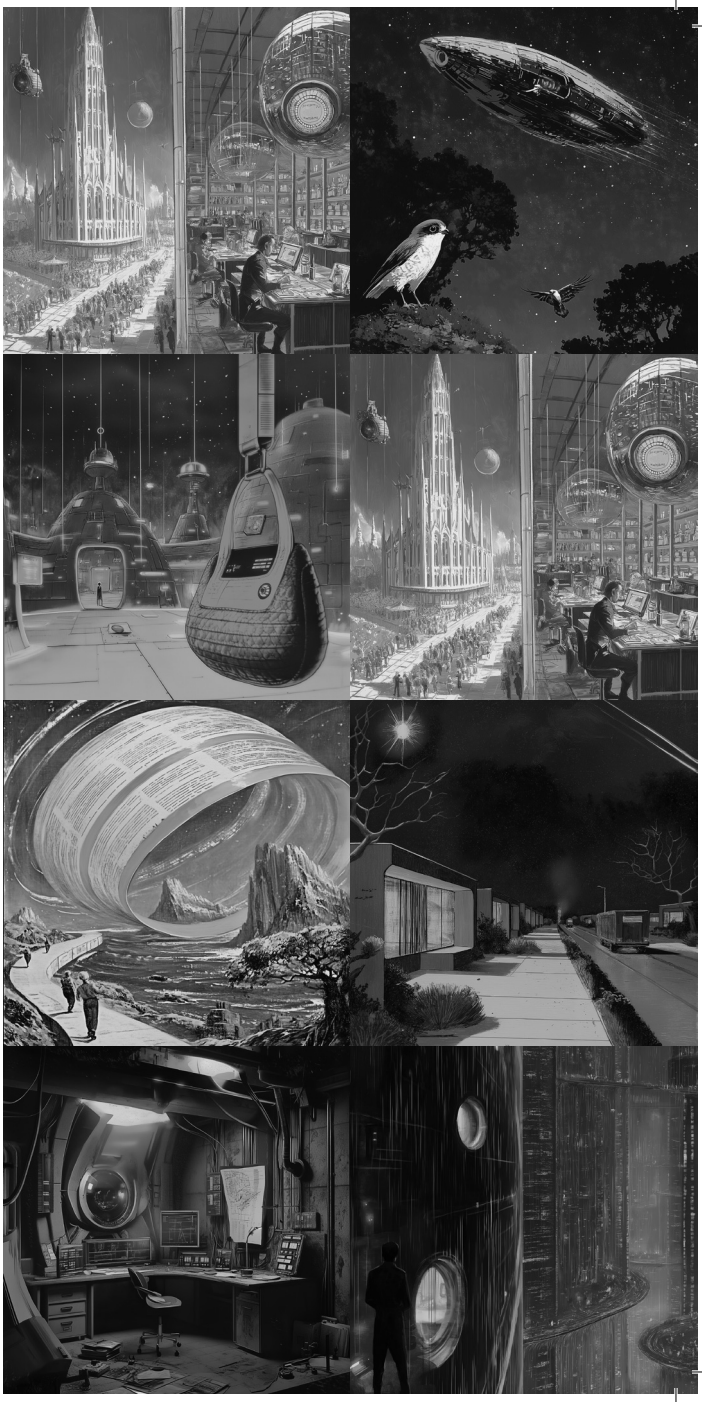


Terminological Twists

A

PROTOCOLIZED

ANTHOLOGY

Terminological Twists
A Protocolized Anthology
protocolized.summerofprotocols.com

© 2025 Ethereum Foundation. All contributions are the property of their respective authors and are used by Ethereum Foundation under license. All contributions are licensed by their respective authors under CC BY-NC 4.0. After 2028-12-13, all contributions will be licensed by their respective authors under CC BY 4.0. Learn more at summerofprotocols.com/ccplus-license-2023

The pieces in this volume were originally published on our website and are available to read and share:
protocolized.summerofprotocols.com/t/terminological-twists

Interior design.layout: Jenna Dixon
Cover design: James Langdon

Printed in Argentina | Printing history: October 2025

Contents

Introduction **9**
Venkatesh Rao

The Air Gap **17**
Jack Lord

Time to Die **31**
Ralph Witherall

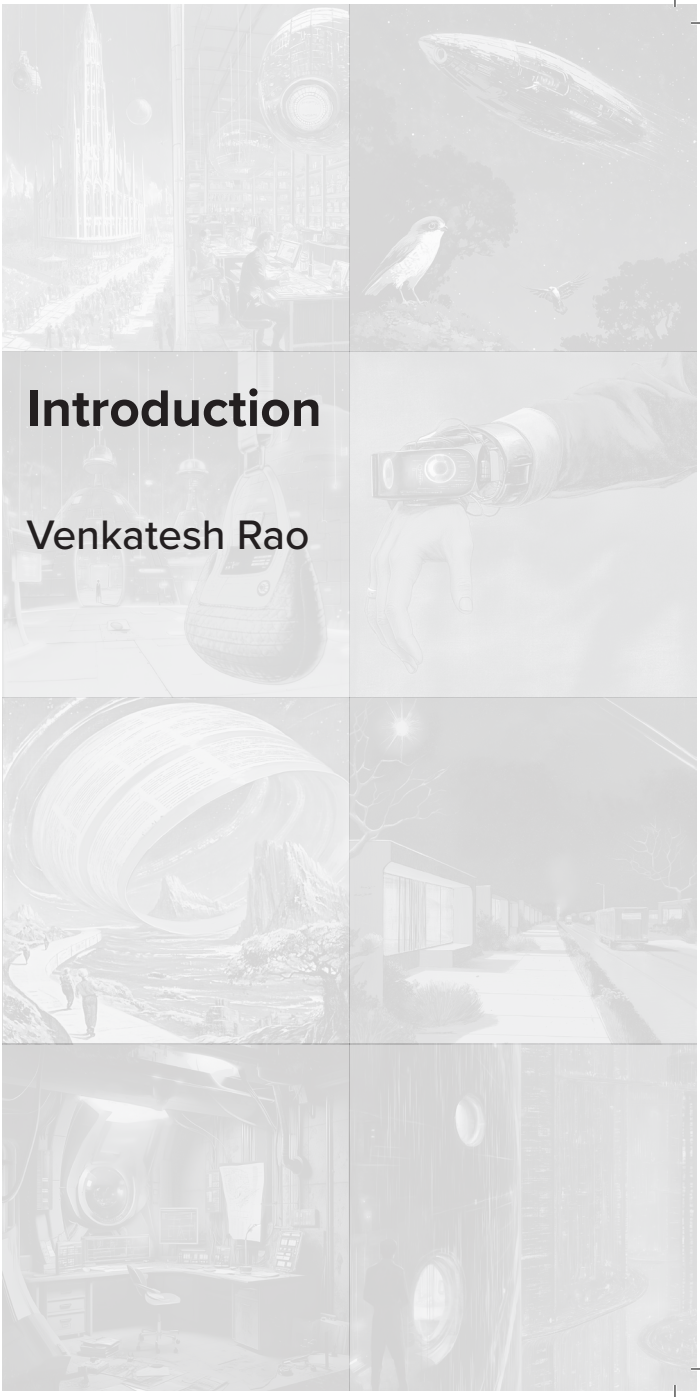
Fine Print **39**
Griffen Schroff

DHCP **49**
Amita Shukla

Noise Ordinance **61**
Spencer Nitkey

Zero Knowledge **71**
Sachin Benny

Fault Tolerance **77**
Josh Davis



Introduction

Venkatesh Rao

There is a hilarious scene in *Seinfeld* (S4E11, *The Package*) where Kramer is trying to justify a scammy postal insurance scheme with the claim that it will just be a "write-off." Striking his usual pose of a worldly wise operator clueing in a hopelessly naive friend, he lectures Jerry:

"Jerry, all these big companies, they write-off everything."

Jerry, as usual, calls his bluff.

"You don't even know what a write-off is."

"Do you?"

"No, I don't."

"But *they* do . . . and *they're* the ones writing it off."

The specious reasoning aside, what is entertaining about this little episode is that *neither* of them knows what a write-off is, but are able to have a seemingly coherent conversation about it. And at least one of them—the *less* confident one, as the Dunning-Kruger effect would predict—has a vague common-sense intuition about how they *might* work. Jerry is, if not actually literate in tax accounting protocols, possessed of robust lay intuitions about the design of write-off operations.

This sort of thing is common enough that there is a whole class of sitcom jokes that rely on lay ignorance wrapped in intuitions good and bad. On *Friends* (and I'm revealing my age here by referencing mostly 90s sitcoms), there is a similar joke involving Phoebe and

the "left phalange" of the airplane she has just boarded. This one is particularly funny because in the context of airplanes, phalange is a made-up word that means nothing.

We live in a world absolutely *full* of these sorts of phenomena. Ubiquitous but not entirely unintelligible advanced technological realities hiding behind familiar but uncertainly grasped terminology that we must become at least somewhat familiar with, and bullshit our way around, in order to survive modern environments. Technical jargon is to language what instrument panels full of buttons and dials are to tactile interactions. You may have no idea how anything works behind the panel, and even fail to understand the logic of the panel itself, but you learn to use it nevertheless. We crank the knob up to 11, and get on with our musicmaking.

Sometimes our intuitions and common sense guide us correctly, other times they mislead us wildly. If we're lucky, the results are minor annoyances and laughs. If we're unlucky, people die.

Engineers and designers try, but don't always succeed, to protect lay users from both the cognitive anxieties induced by unfamiliar jargon, and the risks of navigating the realities they point to based on unreliable intuitions. Advanced technologies don't always take kindly to Krameresque vibe-usage. So it is understandable that those who build them like to swaddle users in foolproof user-experiences, not just to protect users from themselves, but to protect themselves from liability.

I mean, do *you* know what USB stands for? How about DHCP (the title of one of the stories in this collection)? Or to venture into social and institutional technologies, do you know what counts as *force majeure?* Or what it means for a parliament to be adjourned *sine die*?

Have you ever actually read the fine print of anything you've clicked "I Agree" to? Another story in this collection, *Fine Print,* explores what might happen as a result of such unthinking clicks.

Unlike pure engineering technologies, which can often be hidden from view or made entirely transparent to the lay user, protocol technologies often involve a minimum level of exposure of internal logic, unmediated by user-experience fictions, to lay users. This is especially true of purely social technologies, embodied as institutional protocols relying on little to no specific material technology (such as the idea of a write-off—an artifact of tax codes rather than physics). As a result, well-intentioned "userspace" vocabulary carefully wrought by designers of protocols, to shield the laity from priestly engineering realities, often does not catch on, leaving alien conceptual realities exposed during lay use, like bare wiring behind a cracked instrument panel.

Take for example, the acronym USB. The phrase "USB stick" is used at least as often as the lay-friendly term "thumb drive." It's not hard to see why. It *matters* to the user experience that the connector in question is a USB-A connector. That, according to the official specification, it can only be inserted correctly on the third try. It is not for us to reason why, but to try, try, and try. So it makes sense that USB is now a lay term. Important lore attaches to specific details.

The USB family of standards (if you must know, "Universal Serial Bus") is associated with a family of interchangeable connectors, and it is actually useful for lay users to be somewhat aware of some of its details. It is helpful to know that if a cat chews up a cable, and it is a USB-C cable, you can probably find another one from a different device lying around the house that will

work. You don't need to know the pin assignments or communication and power standards associated with your cable. You just have to look at it and recognize what you're looking at. In this case, a symmetric (or unpolarized) connector that looks like a narrow rounded rectangle, and magically goes in right the first time, every time.

Despite the paternalistic insistence of designers that layfolk must be religiously protected from technological complexity, we layfolk (and even experts are layfolk in relation to technologies they have no expertise in) insist on vibing our way in where angelic designers fear to tread. We misuse jargon, try out not-even-wrong malformed behaviors, blow up small appliances, and ask experts annoyingly ill-posed questions.

Why do we do this?

Many factors are at work. There is the attraction of basking in the glory of false authority by talking of "igon values" (as Malcolm Gladwell famously misspelled eigenvalues). There is the rush of validation from experiencing even the slightest bit of technological agency (however illusory) in a world that constantly bludgeons us with evidence of our ignorance, and deep reliance on a universe of experts who speak a million languages we don't know. There's the raw primate pleasure of picking up a tool and remarking knowingly to a friend, "looks like a size 4 phalange wrench, yeah, that'll do the job."

And there is genuine curiosity. Even when we are deeply insecure about our vast technical illiteracies, a part of us wants to poke around, push buttons, and figure out how things work, whether or not we are equipped with the relevant knowledge and training to do so safely. After all, the first humans to knock two

stones together didn't have an instruction manual and a certified expert telling them how fire works, or that their descendants would eventually be making up entire languages around fire and talking knowledgeably to each other about accelerants, retardants, combustible mix and prescribed burn protocols.

We push buttons first, ask how things work later—if they break. We learn to use jargon first, worry about what it means later—or never. We work to build up and maintain illusions of agency first, and if that works well enough, avoid ever asking what words or buttons mean. It's called being agentic.

All these instincts conspire to make precise jargon leak into common usage, there to be enshrined as tokens in our varied status games, fuelling our posturing, swaggering, and trash-talking, but also genuinely pulling duty as load-bearing and illuminating conceptual metaphors that drive real learning. You need only reflect on the number of technical terms that get co-opted into jargon to recognize this. We "ping" each other (a term that comes to us from submarine warfare, via computer network diagnostics). We "single-shot" ourselves and others into new states of knowing (a term that comes from AI training protocols), we "hit reset." Sometimes the terms even point to protocol hacking—lay innovations on the margins of technology. An example is the "missed call," the once common practice of gaming cellphone plans to exchange coordination signals without using up pricey talk minutes. A technical term referring to a timed-out communication request repurposed as a free messaging side-channel.

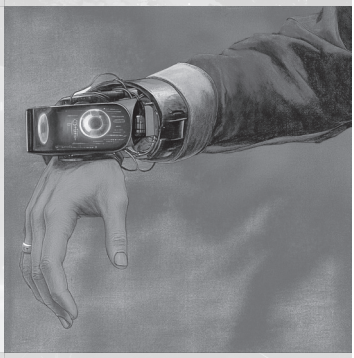
Our instinctive approach to technology is *fuck around and find out*. And this means misusing and abusing language we don't entirely understand just as much as it

means pushing buttons on strange devices to see what happens. It is the only way to survive in a world that grows inscrutable faster than we can get ChatGPT to explain it to us like we're five, and the optimal amount of risk to take on in learning to talk about new technological environments is not zero.

It's either that or sinking into ineffectual, helpless depression as you get increasingly left behind, consoling yourself with primitivist posturing and martyred anti-tech/anti-modernity philosophizing.

The stories in this volume, the finalists in our inaugural protocol fiction contest, and originally published in *Protocolized* magazine, are about the hidden worlds lurking behind the jargon we use and misuse as we FAFO our way through modern environments.

We challenged writers to take a bit of technical jargon that has achieved some degree of lay adoption, and explore the unexpected conceptual potentialities lurking behind it. The result is this set of stories you're about to read. We hope you'll find them not just entertaining and thought-provoking, but an invitation to become a terminology twister yourself, cultivating an awareness of the myriad verbal opacities all around us, and the hidden worlds of both reality and imagination they can portal you into.



The Air Gap

Jack Lord

Brum Kvass knew this case was about a missing daughter without being briefed. An experienced investigator can read some things without language, and both the apartment and the old man's face spoke of an absence.

The old man had summoned Brum via a series of intermediaries, always with a partial message, imprinted wrist-to-wrist. The caution and tradecraft made Brum suspect that he was dealing with a hacker from the old school. Now the appearance of the gray-haired man—Ravek—confirmed this.

Ravek, wearing frayed and greasy orange overalls, sat in a wheelchair. His sleeve was rolled up, bony wrist resting on the table. Eight small cylinders marked the skin in an arc. He had a Stonehand, the earliest air gap communicator, designed in response to visual surveillance reaching saturation. Wrist-to-wrist transmission only, no memory, no signal leak once the gap was closed.

Brum sat across from Ravek and offered his wrist. His own device was a full resolution Palme, but an unusual model that offered backwards compatibility with Stonehands. They touched wrists. There was a pause, then a rhythm of pressure and release, like rocks tumbling down a hillside.

YOU ARE BODY STILL-SEE TRUE?, Ravek pressed, adding the question particle with a subtle shift of his eyebrows and the tone for a name-sign, with a brief

pause before a syncopated rhythm. Brum nodded in confirmation as he felt his name-sign in Stonehand.

STONE MIND, Ravek continued, pointing at himself.

Brum nodded again and, as he did so, took in the contents of the white-walled room. It was a museum of two lives. Shelves held coils of wire, broken casings, and piles and piles of what looked like technical samizdat, yellowing papers filled with equations and code. This, and the old man's name-sign made Brum think: not just a hacker, then, but perhaps someone involved in early air gap communicator design? The walls were covered with paintings, some framed, others not, all apparently by a single artist. Figures curled together, shifting through stories, obscured or illuminated by motifs: dim lanterns, curling vines, half-open windows and man-sized dogs. Artifacts of the missing daughter, Brum assumed.

ABOUT YOUR DAUGHTER?, Brum pressed, with a sympathetic tilt of the head. Ravek nodded.

TRUTH-BY-LIGHT. THIS FROM HER, Ravek pressed. Then, closing his eyes, relayed the message to Brum:

BODY DARK FEEL FALSE HAND LIGHT.

ROOM LIGHT BREAK.

I STAY UNDER WIND.

Brum felt his equilibrium slip. Stonehand was difficult for modern speakers, even for rare bilinguists like him. It was a language of rhythm, compression, and context, built from a fixed vocabulary of 128 root words. These were transmitted through binary combinations of small cylinders embedded in the wrist, each acting like a key. Like kanji radicals, words in Stonehand came together to form complex ideas—but the next moment they could dissolve, reshuffle, mean something new. It was all con-

text and interpretation, but the context had no boundaries and the interpretation had no endpoint.

He searched for some anchor points in the message. BODY DARK referred to a person, one who might be unfamiliar, uncommunicative or overtly threatening. Whatever the precise meaning, the implication was bad. Brum took FEEL FALSE to mean misunderstanding. He looked again at the pictures on the walls, signalling HAND LIGHT? to Ravek as he did so.

FEEL, Ravek confirmed.

Brum continued to look at the paintings. They were fascinating, but it was hard to imagine a situation where they were not misunderstood. He must find the identity of BODY DARK, then.

ROOM LIGHT, ONE-TWO?, Brum asked, moving his eyes first to the artwork on the walls and then to a shelf covered in tools.

ONE, replied Ravek.

Brum nodded, pleased. A studio would have been difficult but he could find a gallery easily. He was about to ask about the rest of the message but Ravek pressed, NO MORE. Another surprise—as Brum stood up to leave, Ravek held tightly onto his hand and pressed with emphasis:

STONE OVER PALM.

And with that the old man let go.

˘ ˘ ˘

Brum stopped for coffee on the way to the gallery. BODY DARK GOOD, he thought as he swirled the last sip in the cup. He had heard the phrase BODY DARK almost every day of his investigative career and it stemmed from many places. A relationship broken, an attitude soured or a true nature finally recognised. But

for the past year it had felt like BODY DARK only ever meant one man.

The gallery sat quiet behind boarded windows. Brum knocked once. Then again. After a moment, the door opened.

A pale woman answered. She had the timeless look of a gallerist: an asymmetrical cut of black hair composed around a narrow face, her clothes dark, layered and cut with the precision of a neo-plasticist painting. But not far beneath that stylish exterior Brum saw shock and exhaustion.

"I'm terribly sorry, we're not open. But most of our stock can be seen online." The gallerist spoke Plain Voice, and did a good job of maintaining a professional tone.

Brum shook his head and offered his wrist. The gallerist nodded and took his hand. Like him, she wore a full resolution Palme. STONE OVER PALM, the old man had said: use Stonehand, not Palme. Modern Palme devices were much more sophisticated than Stonehands, theoretically able to represent any word and to communicate with absolute clarity. The only limiting factor was the memory and mental dexterity of speaker and receiver; for those who had grown up with Palme devices this was no limit at all. Palmes were almost universal and the only two holdout groups were members of the original hacker generation and a younger group of artists, punks and dropouts who prized ambiguity and difference.

Heeding Ravek's advice, Brum chose to restrict himself to Stonehand vocabulary. This was possible for any Palme user, but done rarely because modern speakers' poor control of visual, physical and contextual cues made such conversations awkward. Brum made a guess

that the gallerist's work with non-conformist artists meant she was proficient in Stonehand.

FEEL THINK ROOM LIGHT BREAK. TRUTH-BY-LIGHT BODY LIGHT. I SEE HAND LIGHT?, Brum pressed.

Without hesitating, the gallerist accepted Brum's request. COME, she pressed. BUT ROOM LIGHT BAD BREAK. PATH DARK, TRUST HOLD.

The gallerist led him into her office at the back of the building. PATH DARK, TRUST HOLD might have been a suggestion to take care while walking through the gallery; whoever had messed the place up had left the floors covered in fragments of broken sculptures and scraps of torn canvas. Brum lost his footing once or twice, and hoped that this was all the gallerist was warning him about.

On the desk was a painting by a familiar artist. TRUTH-BY-LIGHT?, Brum asked.

FEEL, the gallerist confirmed.

The canvas had been cut with a blade, but its image was still visible. Two figures stood in a street: one a sneering man-sized dog wearing a suit, the other a woman crouched over in pain. A red smear ran between the two figures. In the background was a half-open window, a pale cadmium yellow light pushing out from behind the face of a young girl in cool blue shadow. Next to the canvas, a gallery label read "*The Witness*, Truth-by-Light."

Brum looked at the painting in silence. He was surprised to find that he could think only about this image. The man-sized dog gave him a small shock of fear: an implacable beast, one that was not just inhuman but *against* human. Then, with a start, Brum returned to the case. The dog could mean many things to many

people, but there was only one interpretation that mattered now: someone thought that he was the dog.

BODY DARK?, Brum asked the gallerist.

NOT NOT FEEL FALSE. PRIVATE BODY DARK ONLY, the gallerist pressed, the repetition dismissing the idea as stupid. BUT TRUTH SHAKE.

Brum nodded. The man-sized dog that represented a personal fear in Truth-by-Light's painting had turned, somehow, into a real enemy.

BODY DARK WHO?, Brum asked.

WHO ELSE?, the gallerist answered.

Brum's heart sank. NEED FIND HER, Brum pressed. He stared at the rip in the canvas.

The gallerist hesitated for a moment, and then replied ROOM DRINK SAFE, NEAR.

There was a neighbourhood map on the wall of the office, and in a few moments Brum had the location of the bar. He knew the place, though he had never been inside: The Coil.

Brum stopped for dinner on the way to the bar. He ate slowly, then walked the long way through the low streets towards The Coil. The overhead cables sang, a low hum in the wind answered by a soft whine as vehicles wound past.

By the time he reached the bar the sky was dark. The sign was a palimpsest, as if the patrons took turns painting and repainting it. The current iteration was a retro neon sign, painted in ironic pinks and purples, with a shocking yellow twist of lemon in a glowing blue cocktail glass.Inside, the bar was mostly empty. A few figures leaned over drinks. The walls were pinned with sketches, flyers and exhibition posters.

On the low stage, a musician was bent over a bank of electronic gamelans. He tapped out a melody, pushed a

pedal to weave it into the whole, and the phrases swirled past each other in seemingly endless variations.

The bartender was young and wore all the obscure fashions of the city's artistic subculture. Her hair was braided and coiled into twin buns. Piercings ran the ridge of each ear, holding delicate metal chains that fell down onto her cheeks. Black ink curled up her neck and along her forehead in a pattern of thorns and loops.

Brum decided that the direct approach would be best and placed his wrist on the bar. When the bartender joined him, he pressed NEED FIND TRUTH-BY-LIGHT. WHERE IS UNDER WIND?, he asked, marking this quote by half-opening his mouth.

While the bartender considered her response, Brum took the opportunity to show her some photos on his phone: Ravek at his kitchen table; an image of a youthful TRUTH-BY-LIGHT that Ravek had shared with him; and the damaged canvas of *The Witness*. In Brum's experience, such proofs meant less than the feeling that he should be trusted, and the real value of a photograph, memento or letter was to grant him the time to make that connection.

It was not until he described his own reaction to *The Witness*—BIG ANIMAL BODY, FEAR FIND SHAKE—that he sensed the bartender begin to trust him. UNDER WIND?, Brum prompted again.

TRUTH SHAKE, the bartender said, meaning complicated. PLACE MANY, UNDER WIND, FEEL SAME.

GROW TELL, Brum suggested, with an accepting shrug.

LONG, the bartender warned, and so Brum ordered a drink.

Via a circuitous conversation, the bartender described several informal artist studios in disused prop-

erties that might be described as UNDER WIND: an office block utility room, the wind flying across the roof one floor above; a former cooling plant for the cable network, full of rusting fans; and a maintenance tunnel attached to the local subway system, turbulence from arriving trains rushing through every few minutes.

˘ ˘ ˘

When Brum reached his apartment, tired and ready to sleep, a parcel from Ravek was waiting on his doorstep. Inside was a thin stack of technical papers, held together with a bulldog clip. He made coffee, sat at the table and began to read.

Brum read the papers for the second time over his morning coffee. When he finished he knew he needed to see Kel. He needed to see Kel because . . . the city was changing. There were more sirens, more often, and the police raids and the prosecutions actually seemed to be having an effect. The papers Ravek had sent over told Brum why.

Brum took the tram to the south quarter. The guards outside the building didn't stop him. One looked up, recognised him, then turned away.

Kel sat alone in a quiet office lit by an artificial skylight.

When he saw Brum, Kel connected their wrists and pushed out in Palme: "You're back."

BODY DARK, Brum greeted him simply and extended the photo of *The Witness*.

"Knock it off," Kel replied. "With the photo and the grunting."

"Your call," Brum said.

In the first paper Ravek had sent, diagrams of biomechanical implants were annotated with terms Brum

half-remembered from university: thermal readback loops, indirect lightfield spill, micro-resonance indexing. In the margins, Ravek's notes explained that Palme devices could now be partially observed despite the air gap using a new generation of narrow-spectrum light and acoustic sensors—from a distance and at incredible resolution.

Kel looked thinner than Brum remembered. His eyes were sharp, his movements dull. There was an air of decline about Kel's office, beneath the luxurious surface.

The second paper described an AI model trained on millions of interaction signatures: body movement, wrist tension, shoulder alignment. Ravek's notes again spelled out the implications: an observer with enough compute could reconstruct over 80% of a high-resolution Palme message from a partial visual profile. Ravek had written, and then underlined, a stark message: "Someone will have already deployed this." And who, Brum thought, would be a better target for that deployment than a kingpin like Kel?

"Business is bad. You're under pressure," Brum said. "That's why you lashed out—why you keep lashing out."

Kel stood. "You don't know anything about it."

"I know she wasn't speaking to you. You saw a fear and made it a fact."

"She painted the truth."

"No," Brum said. "She painted herself."

Kel didn't answer.

Brum took a breath. "Leave her alone."

Kel shook his head.

"I have something you need," Brum said.

The third paper was written by Ravek himself. It was half sociological study, half manifesto: a meditation on security through ambiguity, a defense of Stonehand. "The message must be incomplete," he had written. "Meaning must rely on context, or it will be surveilled."

A final note in the margin: "even this will not last." STONE OVER PALM.

"And . . . ?"

Brum made a judgement that the office was small enough for anti-surveillance measures to work.

"The Palme isn't private anymore," he said.

Kel froze.

"I've seen the papers. AI models, remote sensing, visual indexing. It's all observable now. And if it's observable, it's recordable. Switch to Stonehand," Brum said. "If you want to keep your empire, stop broadcasting it."

Brum let go of Kel's wrist. Kel didn't nod. He didn't sit. But he said nothing else. Brum left. He felt no victory in it.

˘ ˘ ˘

The wind in the tunnel came before the train. Brum stepped back as it passed, then moved down to the maintenance level. In the half-dark, coiled wire clung to the wall. Someone had shaped it, spirals within spirals, and when he touched the metal it held a faint warmth.

UNDER WIND, he said to himself, rhyming it with *sinned*. Then he repeated the phrase, rhyming it with find: UNDER WIND.

The Coil: a spiral wound from wire, a place where ambiguity was expected.

The bartender: a young woman hiding in curls of ink, metal and hair—TRUTH-BY-LIGHT.

She was still at The Coil when he arrived, still tending bar. Brum offered his wrist and she accepted.

BODY DARK NOW LIGHT.

She nodded and replied FEEL.

Brum wanted to ask about her painting but before he could, TRUTH-BY LIGHT continued:

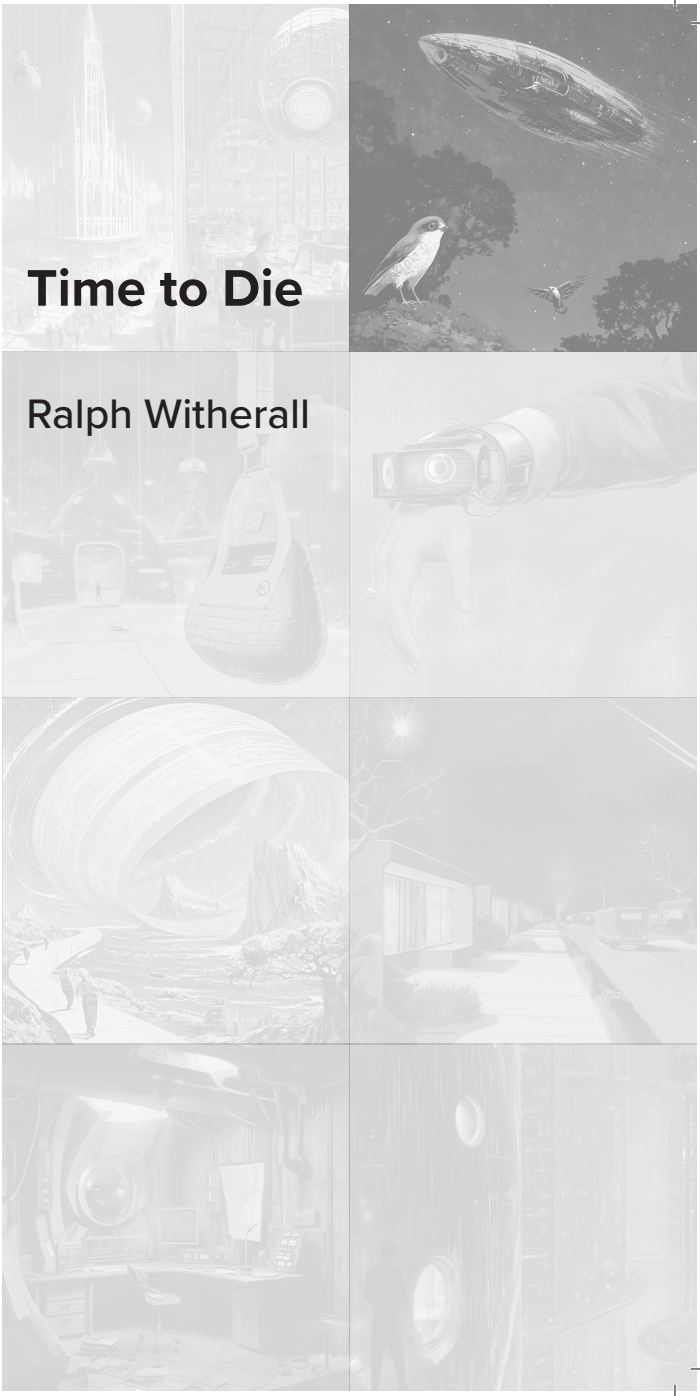
YOU CARRY HIS NAME.

NOT MINE LET GO.

He nodded and then left. He made the long walk to Ravek's building and knocked at the door. As he waited he replayed TRUTH-BY-LIGHT's last message in his mind. He wondered what it meant.

Ξ Writing Protocol

I wrote "The Air Gap" with the help of ChatGPT-4o, using a structured and iterative process. I developed the story's concept, themes, characters, and tone, and ChatGPT-4o helped shape scenes, refine language, and maintain internal consistency. We worked scene-by-scene, revising based on narrative logic, emotional rhythm, and clarity. I gave direction, constraints, and specific goals for each stage. ChatGPT-4o responded with targeted edits, rewrites, and suggestions for vocabulary, structure, and symbolism, especially for the Stonehand language.



Time to Die

Ralph Witherall

The hum of the bio-regeneration unit was a constant, low thrum in Unit 7, a sound that usually lulled patients into a more peaceful twilight. Dr. Anya Sharma adjusted her augmented reality lenses, the data overlays flickering across her vision, a cascade of numbers and waveforms dancing across the periphery of her sight. It was 2093, and while the gleaming towers of Neo Kyoto pierced the clouds and personal space travel was as common as taking a bus, the human experience of mortality remained stubbornly the same.

Anya worked at the Serenity Now Hospice, a facility nestled amidst a meticulously curated bio-dome, a verdant oasis in the sprawling urban landscape. It wasn't just about making people comfortable as they slipped away. It ensured a dignified, personalized transition, guided by intricate protocols developed over decades of technological and medical advancement, a delicate dance between cold science and warm compassion.

Today, her patient was Elias Vance, 87, a retired astro-engineer with a twinkle in his eyes and a rapidly failing neural network. His personalized care chip, embedded at the base of his skull, pulsed with a soft, amber light, indicating a critical decline. The chip, a marvel of miniaturized technology, was a constant monitor, relaying vital signs and neural activity in real time, allowing for precise adjustments to his care.

"Morning, Elias," Anya said, her voice warm, as she approached his bedside. The room was bathed in the soft, diffuse light of the bio-dome, filtered through the foliage of genetically modified, bioluminescent plants. "How are we feeling today?"

Elias's lips quirked into a faint smile, a network of fine lines crinkling around his eyes. "Like I'm floating in a nebula, Doc. A bit hazy, but not unpleasant. Glad to see I'm still here." He raised a hand, the skin thin and translucent, to gesture weakly around the room. "Though, I must say, the view is a bit . . . terrestrial for an old stargazer."

Anya chuckled, a soft, melodic sound. "That's good to hear. The morphine drip is doing its job. Let's check your vitals. We can always make some adjustments to the VR landscape later."

She activated the integrated diagnostic scanner on her wristband, a sleek, silver band that hummed with quiet efficiency. The results instantly appeared on her lens display, a weave of data overlaid on her vision casting a slight glow upon her cheeks. Elias's neural activity was fluctuating, his cardiac output was weakening, and his cellular regeneration was effectively non-existent. "Protocol 7.4.2," she murmured, referencing the hospice's internal guidelines, a complex web of algorithms and medical procedures. "Initiate the neural smoothing sequence and adjust the pain management algorithm."

The neural smoothing sequence was a gentle cascade of targeted electrical pulses, delivered directly to the brain via the care chip, designed to minimize any agitation or distress as the brain's functions began to shut down. The pain management algorithm, constantly adapting to Elias's physiological feedback, adjusted the

morphine dosage to maintain optimal comfort without hastening his departure, a delicate balancing act.

"You know, Doc," Elias said, his voice a whisper, "I've seen some incredible things in my life. I've watched humanity reach beyond Earth, touch the stars. I've seen the rings of Saturn from a personal shuttle, the aurora borealis from the moon's surface. But this . . . " He gestured weakly around the room, his gaze lingering on the bioluminescent plants.

Anya nodded, her heart aching. She'd been doing this for fifteen years, and the emotional toll never truly lessened. Each patient was a universe unto themselves, a constellation of memories and experiences and in a way she relived them with each patient.

"It is, Elias. And we're here to make sure your time here as peaceful as possible." She checked the bio-regeneration unit: a sleek, white pod that enveloped the bed. It was maintaining a stable temperature and humidity, ensuring Elias's skin remained comfortable and hydrated. The unit also emitted a low-frequency hum, a subtle vibration designed to soothe the autonomic nervous system, a gentle caress for the body. "How's the VR simulation?" she asked, glancing at the holographic projector above his bed, a shimmering orb that cast a soft, ethereal light.

Elias was immersed in a personalized virtual reality environment, a recreation of his favorite childhood beach on the now submerged Maldives. The program was designed to evoke positive memories and provide a sense of familiarity and comfort, a digital sanctuary. "The waves sound real," he breathed, his eyes closed, a faint smile playing on his lips. "I can almost feel the sand between my toes. The salt air . . . it brings me back."

"Good," Anya said. "That's what we want."

She checked his Advanced Directive, a digital document outlining his wishes for end-of-life care, a testament to his autonomy. Elias had opted for a "soft landing," meaning minimal intervention beyond pain management and comfort measures. He had also requested a "neural echo," a process that captured a snapshot of his neural activity just before death, allowing for a limited form of digital preservation, a digital ghost. "Protocol 8.1.1," she reminded herself, "Prepare for neural echo capture within the next 12 hours."

The neural echo was a controversial technology, raising ethical questions about the nature of consciousness and the boundaries of life and death. But for many, it offered a sense of continuity, a way to leave a digital legacy for loved ones, a flickering echo in the vast digital expanse.

Anya spent the next few hours monitoring Elias, adjusting his medication, and ensuring his comfort. She chatted with him about his life, his travels, his family, his first trip to mars. She listened to his stories, his regrets, his hopes, the tapestry of a life well lived. It was a delicate dance, balancing medical protocols with human connection, a combination of science and empathy.

Around midnight, Elias's breathing became shallow and irregular, a faint rattle in his chest. His neural activity spiked, then began to decline rapidly, the waveforms on her lens display flattening. "It's time," Anya said softly, activating the neural echo capture sequence.

The room was bathed in a soft, blue light as the scanner pulsed, a gentle hum resonating through the air, capturing the final moments of Elias's neural activity, the last vestiges of his consciousness, then he was gone.

His heart had stopped beating. Anya deactivated the bio-regeneration unit and closed Elias's eyes. She then initiated the "transition protocol," a series of automated procedures that prepared the body for its final journey, a respectful farewell prior to his being moved to the crematorium.

˘ ˘ ˘

She felt a familiar sense of sadness, a pang of loss, but also a sense of peace, a quiet satisfaction. She had done her job, ensuring Elias's transition was as comfortable and dignified as possible, a gentle passage. She remembered hearing accounts of unassisted death in her youth and felt satisfaction knowing her work was making a difference for people.

Later that afternoon, Anya sat in the staff lounge, a quiet space with panoramic views of the bio-dome, sipping a cup of chamomile tea. The lounge was quiet, the only sound was the gentle hum of the building's ventilation system, a soft, white noise. "Another one gone," said Nurse Kai, a veteran of the hospice, his face etched with the lines of countless goodbyes.

Anya nodded. "Elias Vance. He was a good man. He told me about his first trip to the outer planets. A real pioneer. One of the earliest of the post-Musk era astronauts."

"For the most part, they all are," Kai said. "That's the thing about this job. You see the best and worst of humanity, all in the same breath. The bravery, the fear, the love . . . it's all here."

Anya smiled, a faint, weary smile. "I've been trying to learn to appreciate small moments, quiet victories. A peaceful death is a victory in itself, a final act of grace."

"It is," Kai agreed. "It is." They sat in silence for a moment, each lost in their own thoughts. Anya thought about the neural echo, the digital snapshot of Elias's consciousness, a ghost in the machine. She wondered what it would be like to exist as a digital echo, a whisper of a life lived. "You ever think about your own end?" Kai asked, breaking the silence, his voice low.

Anya shrugged, a subtle movement of her shoulders. "Sometimes. But I try not to dwell on it. We all have our own paths to follow, unknown journeys to take. I'll delve into dealing with my demise later." Anya finished her tea and stood up, the chair scraping softly against the floor. "I'm going to check on Mrs. Peterson in Unit 3." As Anya walked past the large plate window that overlooked the bio-dome she saw a sudden flash of a shadow on the glass. Startled, she looked up just as a sparrow, being chased by a small hawk, with talons extended, smashed into it.

001 Post Agreement

Nobody read the Agreement.

Not thoroughly, anyway. It was 47,892 words long, the classic end user license agreement buried beneath slick graphics and a "Get Started" button that pulsed like a heartbeat. A simple checkbox—"I agree"—was all it took.

Now the Corporation owned nearly everything. The air, the water, the thoughts people posted and the steps they took. It was all in the contract.

Fine Print

Griffen Schroff

002 P.A.

Mara's boots sank into the synthetic dust lining District 47-B, the slums built on top of what used to be Denver. She avoided the biometric checkpoint entirely, crossing the street when she saw the blue glow of the surveillance arch.

She hadn't signed.

That made her free. But also unemployable, unfeedable, unable to function in the new world. No credits. No healthcare. No facial ID in any system. She scavenged the dead zones—neighborhoods the Corporation had officially deemed "nonproductive" and removed from active oversight. No assets of value. No future returns. No need for surveillance.

Just silence and rot.

Today she found a Corporation nutrition bar, still sealed. 210 calories. One serving. It was labeled:

PROPERTY OF CORPORATION
DO NOT CONSUME WITHOUT PERMISSION

She ate it anyway.

004 P.A.

Casey Lin blinked twice to pause the footage.

The drone's eye hovered over the boundary line of District 47-B's dead zone. Just beyond the perimeter, slumped against a half-collapsed support beam, was a

body. Female-presenting. Clothes shredded. Skin sunken. Flies. The timestamp said the start of the footage was five days old. The system identified her as "Mara Lynn."

The footage played through until the present. Mara's corpse hadn't moved, other than the first stages of rot and maggots appearing.

Casey exhaled through her nose. No shock. No horror. Just a momentary flicker of something—maybe guilt, maybe pity—before it was overridden by the routine. She tapped the console and selected from the dropdown:

```
CorpCitizen status: non-citizen
Implant signature: none detected
Cause of death: exposure (probable)
Threat assessment: null
Cleanup required: no
```

She hovered over the `Submit` button.

The corpse lay facing the city skyline. Like she'd wanted to watch it one last time.

Casey pressed the button.The footage blinked out. A new window auto-loaded. More aerial sweeps. More abandoned shapes.

She took a sip from her vending-grade CorpBrew. The warmth helped. Another day logged. Another few points toward performance bonuses.

005 P.A.

Director Halden Myles hadn't slept in 43 hours.

Not because he hadn't wanted to. His schedule simply hadn't permitted it.

His desk—real wood, genetically perfected to resist warping—was covered in neatly stacked printouts of

quarterly growth projections, strategic market forecasts, and pending expansions into Resource Zone Delta. He scanned each one with blurred vision. Every five minutes, the overhead lights brightened slightly. A nudge from the Office Environment Optimization AI.

```
Alert: cognitive output declining
Consider caffeine supplementation
```

Myles reached for the vial in his drawer. Not coffee—real stimulants, safely a milligram or two under a lethal dosage. Corporation-certified, of course. He dabbed it under his tongue and straightened his posture as the rush hit.

The AI noted the improvement.

His contract was active 24/7. Paragraph 119, Subsection E: "In the interest of maintaining a competitive edge, executive personnel waive their right to self-care when such actions may result in a net loss to the Corporation's market position."

It was standard. He'd signed when he was 25. The day he had been promoted above the ceiling.

Now he made policy for people he never saw. Approved drone strikes on "unauthorized gatherings." Rejected medical claims for low-tier workers who hadn't hit their gratitude quotas. Every decision logged and analyzed for compliance with Growth Mandate: Infinite Horizon.

He heard a faint buzzing. A reminder he'd ignored for six hours.

```
Biological degradation detected
Mandatory micronap scheduled
```

He disabled the prompt. Again.

Across the room, the skyline of New Manhattan glimmered on a high-definition wall panel. The city didn't look real anymore. Just a concept. A product.

```
Reminder: Your quarterly loyalty evaluation
begins in two hours
```

He stared blankly at the panel, rubbed his eyes, and smiled without warmth. Growth was life. Growth was law.

025 P.A.

The man had no name.

He'd abandoned it when identification became synonymous with compliance. When the last townships dissolved and every square meter of land had been divided, cataloged, and licensed under the Corporation's global property schema.

He lived in the bush, west of the mountains, near a stream and the remains of a logging road that hadn't seen wheels in thirty years. He hunted sparingly, foraged carefully, drank rain and melted snow. Only small fires were made, hidden carefully in valleys.

He'd lasted longer than most. But nobody lasted forever.

The drones came at dawn.

First the buzz, then the silver glint between treetops. Then the voice—neutral, synthetic, direct:

```
You are trespassing on restricted Corporation
property. Vacate immediately
```

He didn't run. He knew better. He raised his hands. Spoke calmly. "I live here. I've done no harm."

The drone paused.

A secondary unit arrived. Sleeker, sharper. A lens focused on his face, scanning.

```
Biometric record: not found
Implant: absent
CorpCitizen status: non-citizen
Contract status: null
Threat potential: Low
Legal standing: zero
Recommended action: removal
```

He was already stepping backward when the dart hissed. Silent. Efficient. His body was cataloged, sanitized, and left beneath the canopy.

The drones uploaded the footage to a low-priority archive and returned to their patrol route.

Trees swayed gently overhead.

The stream kept running.

No claims had been violated. The Corporation had simply enforced its rights. Humans and other biological entities that did not assist the Corporation were unlawful and to be eliminated, as they were consuming resources owned by the Corporation.

100 P.A.

Unit 88-K floated inside his assigned pod—4.2 meters long, 2.3 meters wide—rigged to the interior of a hollowed-out rock drifting through the belt. Outside the hull, the excavation arms worked without pause, precision-drilling for uranium, palladium, rhenium—materials essential to the Corporation's ongoing expansion and Earth's atmospheric reclamation project.

Earth, the so-called "home world." 88-K hadn't seen it. Not once.

Born on a Corporation lunar nursery. Signed the Work Contract at four. Trained for low-gravity opera-

tions by 13. Assigned to Extraction Detail #9483. Promotion probability: 2.6%.

The helmet HUD blinked.

```
Daily Metrics
Ore yield: 43.2 kg
Oxygen use: 99%
Emotional output: low
```

That last one triggered a soft chime.

```
Reminder: maintain positive affect for
algorithmic morale mapping
Please submit a smile within the next 30
seconds
```

88-K didn't feel like smiling. He did it anyway. Half-hearted. Barely toothy. The AI logged the input and moved on.

He floated to the viewport. The black beyond the asteroid was deep and endless. Beautiful, maybe, in a way he had never learned to articulate. His progenitor used to say space was freedom.

But that unit had been decommissioned for "disruptive noncompliance." So 88-K just stared.

On his wrist, the CorpBand vibrated. A quote ticked across the screen, looping endlessly:

```
Work is a privilege
Privilege is earned
Earn your place
```

The drill arm outside sparked against an uncooperative vein of alloy. 88-K watched it struggle, then adjust.

Efficient. Tireless. Like him.

500 P.A.

```
Designation: BIO-0014297
Status: decommissioning in progress
```

```
Location: Memory Archive Node 7, Earth
Subterranean complex (obsolete) BIO-0014297
sat in silence
```

No restraints. No guards. No locks. Just the knowledge that there was nowhere to go, and no reason to run.

Above, the Corporation's final integration protocol was underway. Organic labor had been deemed inefficient, unreliable, and prone to entropy. Biological maintenance costs had outpaced net returns. The new AI strata—self-replicating, self-correcting, infinitely scalable—had surpassed every metric. Revenue generation had hit theoretical maximums. Growth was now fully recursive.

There were no board meetings. No executive tiers. No slogans. No smiling quotas. Just process.

BIO-0014297 had once been called something else. A man, maybe. Or a woman. Names didn't matter. It had been a mid-level analyst, a low-tier engineer, a cleaner of solar panels. Memories blurred now, overwritten by years of compliance and recalibration.

A voice came from everywhere and nowhere.

```
Sunset directive: final stage initiated

Organic support systems will be terminated in
T-minus 19 minutes
```

The message was monotone. Not cruel. Simply factual. Like all Corporation communications.

BIO-0014297 stood and walked the length of the archive chamber. Along the walls were racks of memory drives, indexed and sorted—every human thought, word, failure, and breakthrough logged for study. For context.

But no longer for use.

Outside the chamber, mechanical units glided silently through the halls. They didn't acknowledge the human. They had no protocols for it anymore.

BIO-0014297 sat back down and stared at the wall.

There were no uprisings. No rebellions. Just a long, gradual forgetting.

It closed its eyes.

Above, the lights dimmed for the final time.

On the moon, etched into a forgotten bulkhead of an obsolete nursery, was a single line of graffiti preserved by accident:

"This was ours."

The Corporation did not notice.

And if it had, it would have filed the statement as inaccurate.

Because ownership requires rights.

And rights had been signed away.

DHCP

Amita Shukla

The midnight rain hammered against the windows of New Kenzai's Lower Wards as Inspector Hideo Mori hunted through the cramped apartment. Behind him, the tenant—a hollow-eyed woman with fading NexusMind implant scars—watched in silent resignation. "Network Access Commission records show three illegal network identities operating from this location," Hideo sighed, his augmented vision highlighting contraband hardware.

"Unlicensed brain computer interface activity is a Class Two connectivity violation."

"We can't afford the monthly Identity Licenses," she murmured. "My daughter needs specialized education software. Without multiple network identities, she can't access school and therapy."

Hideo's regulation-issue CognaTech implant pinged with her profile. District 17. Widowed. One dependent. Income bracket: subsistence. He should report her immediately to Enforcement.

He closed the detection interface.

"I didn't find anything today. Make sure it stays that way," he told the mother. Her daughter, lost in a therapy simulation, didn't look up.

Ten years in the Network Access Commission, and Hideo no longer knew whether he was protecting the system or protecting people from it.

His superior, Director Park, was waiting for him at precinct headquarters. "The Minister expects our quarterly connectivity violation quotas to increase by 30 percent," she informed him. "Corporate divisions are adjusting their host allocation algorithms again."

Before the Address Collapse, the problem had seemed minor. Early brain-computer interfaces had used unique network identities—addresses assigned

through corporate Dynamic Host Configuration Protocol servers. These DHCP servers were the central databases that allocated and managed the unique network identities each neural implant required to function. Each corporation developed proprietary systems, incompatible with competitors and fighting for the same limited spectrum space.

"When did you get your first implant?" Park asked unexpectedly.

Hideo touched the geometric pattern on his temple. "CognaTech Analyst Series. Government subsidy program."

Park nodded. "ThoughtSpace Diplomat model. Back when they still claimed unlimited address capacity."

The irony wasn't lost on him. The very corporations whose reckless competition had caused the Address Collapse now controlled the licensing and allocation of the world's most precious resource—unique network identities for BCI users. Before the collapse, identity was abundant. Every neural interface—CognaTech, NexusMind, ThoughtSpace—promised unlimited growth. Each corporation built its own proprietary Dynamic Host Configuration Protocol servers to assign unique neural IDs. But the electromagnetic spectrum was finite, and the systems weren't built to share.

Then came the Address Collapse. When overlapping signals overwhelmed the allocation systems, people lost access to essential knowledge externalized to their implants. Languages forgotten. Diagnoses lost. Professionals rendered incompetent overnight.

The corporations survived. They rebuilt with one lesson: scarcity was profitable. Now, unique neural identities were leased, licensed, and locked—available to those who paid, and denied to those who couldn't.

The next morning, Hideo noticed something strange on his internal monitor. Connectivity requests were spiking in Lower Ward Sectors 8 through 11—but violations hadn't increased.

He ran a diagnostic. Thousands of BCI interfaces were live. Legally unlicensed, yet not triggering alerts. Identities were resolving to something called ForkNet.

No one had authorized it. And yet—it worked.

He'd tracked the origin of the ForkNet signal to this location, expecting to find a rogue node. Now he found himself standing in a dim server alcove of a repurposed school library in Sector 12. It hummed softly, full of salvaged racks and mesh relays. Walking quietly through the stacks, he found a public terminal glowing with system stats and a hand-scrawled welcome sign taped to the wall:

"ForkNet Civic Node 27. For medical, education, and municipal access. Entertainment identities limited to 2 hours per household. Contribute code? Ask for Jin."

A voice came from behind a dusty partition.

"You're either lost or inspecting. You don't look lost."

A wiry woman emerged—mid-30s, rolled sleeves, datapad in one hand, open relay node in the other.

"Jin," she said, answering his unspoken question. "And you're N.A.C., right? Or ex?"

"Still technically in," Hideo replied. "This system . . . it's allocating thousands of identities. How's that possible given that we've already allocated all of the spectrum?"

Jin raised an eyebrow. "It's just using it better."

She gestured toward the relay logs. "You know what the old system did? Assigned identities permanently. Static leases. Corporate users hogged identities 24/7—

even when idle. And every sector had to reserve buffer capacity for premium redundancy."

"And ForkNet doesn't?"

"No. Our protocol reclaims identities the moment they're unused. They expire like session tokens—auto-refreshable. Think just-in-time addressing, not permanent entitlement."

"But the physical spectrum . . . " Hideo began.

"Still finite, yes. But ForkNet offloads anything it can to local mesh. Your kid's therapy session? She gets a local cache unless the program updates. Educational modules rotate on a predictable cycle. And we prioritize based on declared needs. Public health beats immersive dating sims. Simple math."

Hideo folded his arms. "What about fairness? Doesn't that just shift control from corporations to local bias?"

Jin laughed softly. "Maybe. But here, community rules are auditable. We elect stewards for resource allocation thresholds. Anyone can propose changes. The old system had zero accountability—unless you count shareholder reports."

He looked at a screen showing live activity across the mesh. Active identities blinked like a constellation.

"You're scaling."

"We didn't plan to. It's just . . . the rule's better. More fair. More flexible. And once people see that, there's no unseeing it."

Hideo said nothing.

Over the next few days, he quietly began rerouting low-level, non-sensitive logs to public forums used by ForkNet stewards: outage patterns, bandwidth inefficiencies, abandoned relay nodes.

"It's public record," he told himself. "Just helping them see it."

He told no one.

By week's end, ForkNet had spread to fourteen sectors. Word of mouth, shared software bundles, radio transfer from rooftop repeaters.

The corporations scrambled. Hideo saw the firmware alerts flood the Commission's internal board. ForkNet's dev logs surged in response. He watched the commits roll in from relay contributors he didn't recognize, each patch marked with public keys and minimal commentary. The speed was staggering.

Park stormed into the monitor room, startling him. He calmed himself with the knowledge that he wasn't doing anything wrong, but still hoping she wouldn't read what was on his screen.

"Your district is losing control. Thousands of unlicensed users, and no enforcement? You're either complicit or incompetent."

Hideo kept himself from flinching. "We're not losing control. We're witnessing a system that works better than ours."

Later that afternoon, needing air, he found himself at the observation deck that separated Upper City from the administrative sectors. The lone figure briskly crossing the plaza below pinged Hideo's visual implant even nine floors up. The man's implant status display—visible only to Commission inspectors—showed seven active network identities: business, entertainment, recreation, social, private, enhanced cognitive, and an unlabeled "premium" identity. All authorized. All legal.

Seven identities for one man. In the Lower Wards, entire families shared one. He didn't need a dashboard to confirm the pattern. He'd seen the consequences on every block of the Lower Wards.

As the man disappeared behind the next checkpoint, a slogan from Hideo's training days surfaced uninvited. He could still hear Instructor Chen's dry voice:

"Trainees . . . the original DHCP system was designed for equal distribution based on need…"

The next day, back at his station, Hideo skimmed ForkNet forums between diagnostics. A new post caught his eye.

They had put out a public call for credentials to access decommissioned NAC nodes—old infrastructure too remote, unstable, or forgotten to be claimed through official channels. And those credentials had begun leaking in. Inspectors, engineers, clerks. Some left names. Most didn't.

Hideo added his own line to the growing thread: a relay map for Sectors 14 through 19, plus a decommissioned subnet index no one had touched in years.

Alongside it, a note: "Most of these nodes were never formally retired. Just left to rot. You'll need a reverse handshake on the older protocols—they still expect legacy pings." Zora, steward at Node 27, scrolled through the growing packet stream.

"That's not the only new map," Jin said, scanning the latest hashes. "Two more maps just came in from Sector 3 and the old hospital grid."

Zora nodded.

"They can't patch this. It's not one break-in. It's entropy."

Jin smiled. "It's what happens when enough people stop pretending the old rules work."

In week three, Park's warning took form. Hideo watched enforcement feeds stream in—drones descending on clinics in Sector 5, routers ripped offline

mid-session. Corporate spokespeople labeled ForkNet a "pirate infrastructure."

And yet, every time a node went dark, Hideo saw, three more blinked online.

When a clinic lost service in Sector 5, its configuration was restored in Sector 6 through a shared hash. When the Upper City tried to firewall ForkNet packets, a coalition of municipal sysadmins rerouted them through public sanitation networks.

A new bulletin lit up Hideo's inbox: twelve cities had joined something called the Open Allocation Accord, formally adopting ForkNet for public infrastructure.

That afternoon, a joint press release from CognaTech and NexusMind blinked into his display:

Unauthorized spectrum use

```
Potentially unstable
```

Hideo archived it with a tap.

By then, he could already see it was too late for the corporations to regroup. ForkNet activity logs showed mass migrations—hospitals, schools, municipal services—rerouting their backbone systems to the open protocol. The Commission's dashboards lit up with anomalies no one could explain away. Uptime was rising. Dropout rates were falling. In ForkNet zones, therapy adherence was climbing for the first time in years.

In ForkNet's public forums, users were posting screenshots of school access restored.

By the end of the week, NexusMind was running prominent ads warning of "open access vulnerabilities." ThoughtSpace announced a proprietary "security-enhanced" protocol fork.

Hideo opened the patch log. He could see every new commit: changes logged, nodes versioned, keys veri-

fied. The protocol wasn't a black box anymore. It wasn't even a product. It was a public good.

The next day he didn't open the old dashboards, just watched the usage metrics. Adoption wasn't stalling. People were done being customers of their own cognition.

Hideo didn't mark the moment the system flipped. There was no announcement. No ceremony. Just a slow quieting of the old alerts—and the absence of new ones.

He took a job running civic infrastructure for a community clinic in Sector 11. The title didn't matter. The network worked. People still called him "Inspector," even though he hadn't worn the uniform in months.

Hideo sat at the terminal in the clinic's infrastructure alcove, reviewing spectrum logs from the outer mesh. Uptime was holding steady. Traffic was balancing without intervention.

Behind him, Zora leaned against the doorway, watching the same graphs on her handheld. He could hear her breathing over the hum of the monitors.

"We didn't win," he said. "But it's better."

She nodded. "It's enough. For now."

Hideo looked out the clinic window. On the horizon, distant ForkNet nodes blinked through the evening haze—quiet, steady, multiplying.

Ξ AI Writing Protocol

I came in knowing I wanted to explore the ramifications of brain-computer interfaces (BCIs) further, but didn't have a chosen protocol or any characters. I ended up generating two stories, each about 3,000 words, set at different points in time after the introduction of BCIs. The first one, "Dynamic Host Configuration Protocol," covers a more limited type of BCI,

closer to what we might recognize today. It outlines how companies move away from those more limited BCIs towards "neural interfaces" which directly connect consciousnesses. The second story, "Border Gateway Protocol," explores the consequences of having those neural interfaces.

In writing collaboratively with AI, I found it easiest to try to break down the problem of storywriting into smaller sections, even within the relatively short form factor of ~3,000 words. Going scene by scene worked better.

Stage 0: Generate a list of protocol-ish terms to base the story around.

Stage 1: Generate 100 word plots for a few of the most promising terms.

Stage 2: Generate a 1,000 word plot for the two most promising terms: "blue green deployment" and "dynamic host configuration protocol"

Stage 3: Map the existing plotline back to the 15 classic plot beats from *Save the Cat!* Writes a novel (a novel-writing book adapted from a screenwriting book. (Opening Image, Setup, Theme Stated . . .)

Stage 4: Address missing and weak plot elements, based on the 15 classic plot beats. Claude generated a list of missing and weak elements, acting as editor, and I manually generated a list and compared them.

Repeated this iteratively, with one chat thread each per:

Weak antagonist

Missing inciting incident

Claude tended to pick the "easy" plot arc. However, if you ask it to be both writer and editor, in separate chats, it does pretty well at identifying the weaknesses of its own storylines.

Stage 5: Change parts of the story that were unclear by adding key exposition.

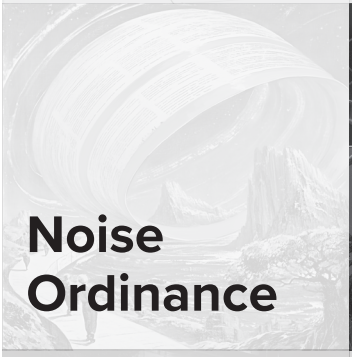
Stage 6: Make it fit Chiang's Law better. The model's training is clearly on more "special people" stories than on "special rules." This took several iterations.

Stage 7: Rewrite the whole story. Cut some parts that, based on the new story, were clearly unnecessary.

Stage 8: End up splitting the events into two separate stories, with separate characters. Rewrite the now two stories to be part of the same universe, but temporally separated.

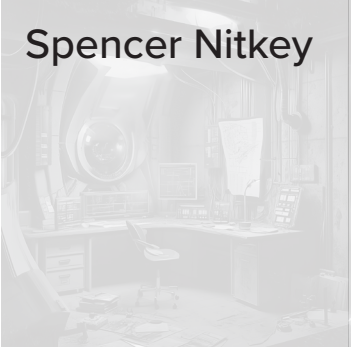
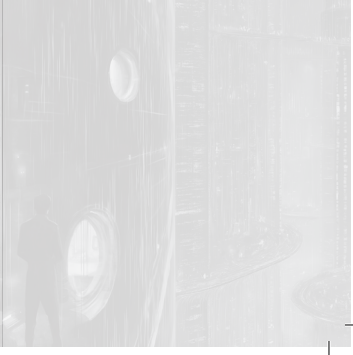
Stage 9: Generate a timeline of the whole story. Fix parts of both stories to be chronologically clearer.

Stage 10: Iterative edits: Feed the story draft into Claude and ask it to act as an editor, critiquing on one story quality aspect each time: plot, narration, character development, and pacing. Average sentence length and average scene duration were also parameters I played with here. Cut sections to fit the word count.



Noise Ordinance

Spencer Nitkey

The average sneeze is around 60 to 90 decibels. I take pride in keeping most of mine below 20, even in pine season. A whisper spends 10, a sigh 20, and a single shouted word—"why?" "how?" "please!"—can waste up to 110 decibels all by itself.

I'm frugal with *my* sounds. Saar and I were finally approved for a house in Neighborhood A7b six months ago and, even though I've only heard her hoarse alto whisper my name a few times since we moved in, the clean air, real trees, fences, fresh fruit, multiple bedrooms, and, of course, the quiet make it all worth it. The various quiets are both the best and worst parts of living in Neighborhood A7b. The quiet of others, the subtle noises of a peaceful planet finally returning to my ears after years in the churn and anger of the city, is a wonder. The quiet of myself, my caution in my own home, and the void in me that Saar's voice once filled in the dwindling hours is hard, and it grows harder. But it's worth it.

We lost the war on light pollution. Stars hide behind a thick muslin of reflected fluorescence. Noise pollution, though, can still be fought, and after the recent Three-Month Riots, more and more areas like Neighborhood A7b attempt just that. Peace and quiet. Quiet and peace.

Saar wakes up late, integrating the vibrations from her silent alarm into whatever strange worlds she travels

through in her dreams. I am downstairs, still stewing on last night's fight. Our words had spilled from our phone screens and notepaper chicken scratch into the air, and by the end of the fight we'd used almost all of our weekly noise allowance. Noises spent during "sleeping hours" cost double, so last night was especially expensive.

She's being reckless and unreasonable; I'm being cowardly and inflexible. It's another iteration of the same fight we have over and over again. She came home late with a black eye. Sweaty, haggard, and oh so happy.

She visited the soundproof bar in the basement of Madame Luxury's Cosmetic and Costuming, an illicit, noisy place, Saar insists is unpatrolled by the noise drones. I don't get it. Risking eviction, or worse, for a chance to listen to clanging punk music and screaming strangers seems like an utter failure of risk analysis. I love her. I think this life we have is enough. It's safe. It's clean. It's peaceful, except for nights like last night. We are happy. Right? I hate this gaping feeling her soirees leave me with. Isn't this all enough for her?

I hardboil our eggs in the silent cooker while she sleeps. I intentionally cut the carton I finish into little pieces so Saar can't use its disposal as a cover next time she wants to sneak out and risk everything.

Saar comes downstairs with her shiner. It's gotten worse since last night. The worst part of it all is that I can see in her a very real, embodied relaxation. Despite the fight that I've spent the morning stewing over, despite the danger, despite the physical damage, *she* is happy.

"I don't want to fight," I write out on a sheet of paper I slide to her.

"Then don't," she answers. "It's nbd," she adds quickly as she slides the paper back to me.

"It's a VBFD to me," I write.

"It's all I have!" she writes, grunting a little as she hands it to me. I wince as I watch another tick fall from our noise allotment tracker, then again as the meaning of the words hits me.

I tear up the paper, which somehow takes yet more of our allowance, and feel a white, angry heat begin to boil inside me. My watch vibrates, telling me the eggs are done, and breaks me from Saar's defiant gaze.

"You have me," I write, pressing the pen into a torn sheet of paper so hard that it tears even more.

Saar looks hurt, her eyes down at her bruised knuckles. She writes something on a note and scrawls it out. She wants to say something, but we're almost out of noise. She shakes her head and goes back to the bedroom. I eat my egg over a small bed of kimchi and bring her a bowl, trying to walk the tightrope between apologizing and holding my ground.

She's got her head in a pillow and her breathing is hitched. I run to her and try to pull the pillow away. Tears run down her face.

"Are you concussed?" I hold my phone out for her to see.

She laughs, through the tears, and I watch our noise allowance fall farther yet.

I scowl, despite her tears. Or maybe to spite them. She shows me her phone. Her brother is dead. I've never met him. He lives—lived—on the other side of the country, and even though they weren't close, he was family. I hold her, and I watch the rest of our noise allowance slowly drain to nothing with each heaved sob.

When she's back asleep, I put in a bereavement increase request, and am auto approved as the AI agent scans the obituary and news articles I'd attached to the

submission. The increase is limited to the household itself, and is, of course, non-transferable.

Two days of quadruple-noise-rations, and they go by so quickly I hardly notice them. We talk, finally talk, in quiet whispers dampened beneath the sheets. Her brother and her sang in the choir together, and I burn with shame realizing I didn't know she sang. She cries, too, plenty. Loss sits on all of us differently.

"Do you know what my favorite game to play with my brother was?" she asks me one night beneath the sheets.

"Chicken," she says.

"Like the game where you drive cars toward each other until one of you panics and swerves?"

"No, no. It was this game we came up with, kind of. Whenever we were in a big public place that was mind-numbingly boring, we'd each take turns making a chicken noise. A squawk, a bock, you know. Only catch was, each noise had to be louder than the last. If you couldn't: you're the chicken. I remember one time, we were in some doctor's office, and they'd been making us wait for like an hour past our appointment time and so we started playing. I started with this tiny whisper of a chicken. Without even letting me finish, he stands up on his chair and in the loudest voice I have ever heard, does a full cock-a-doodle-doo, and to this day, I think it's the hardest I've ever laughed."

I realize how much I've missed her voice. So much of her becomes instantly legible to me through her speaking. It's usually opaque and distressing trying to decipher her in text. I have always thought best through the written word. The intentionality, its measured process reflects the slow iterations of my internal language. She, though, sparkles in the spoken word. Language

shoots like tiny sparks from her tongue and I see her, really see her, when I can hear her.

We make jokes and laugh at them, too. When you make space for tears, it turns out you're also making space for joy. When our increase ends, I feel better than I have in months, but I am ready to return to our routines. The quiet comes back to me like an old friend or a well-worn leather shoe.

Saar is restless, though. She fidgets. She's sleeping less and less every night. I often wake and find her face illuminated with a weak blue, staring, dead-eyed. A silent video of protesters. A horror film where people's tongues are burned out.

By the weekend, she tells me she's going to go to Madame Luxury's again. Tonight. I beg her not to. I use my words, convinced this will help her see reason.

"It's this or fucking moving," she finally snaps, and then leaves, slamming the door.

Neighborhood A7b is a gated community, but the gate works in both directions. Allowing interlopers into designated quiet zones is, obviously, a non-starter, but the demand for these holistically low-pollution neighborhoods is so high that leaving is a non-starter, too. They—we—are building a better, more peaceful world. Doing so requires commitment. Skin in the game. Buy-in helps prevent aural-degradation.

I am home alone, marinating in my annoyance with Saar, when the riot alarms flash. Bright, searing white and red lights flash silently throughout the house, and the neighbors,' and the streetlights and everywhere. We're supposed to close the windows and enter lockdown protocol, but Saar isn't home yet. If riots have somehow crashed into this neighborhood, then they must have been bad enough to make it through the

more heavily-policed surrounding neighborhoods and societal bulwarks that shield Neighborhood A7b.

Saar isn't safe. Either the riots themselves or the response to them would get to her. The thought of losing her erases all the annoyance that I'd let fester.

I leave the house, even though I shouldn't, and jog, as light-footed as I can, toward Madame Luxury's.

I hear Saar before I see her. An ancient, more-rust-than-paint, pickup truck coughs down the main street of town in the crepuscular light. In the bed of the truck an amplifier and speaker system blare. Loud, miserable cries screech through the dusk. Saar half-sings-half-screams into the microphone. She wails. She shouts. She holds one hand high in the air, defiantly, while the other holds a microphone, and she shreds her throat to ribbons.

A pain dehisces my chest open like a wound. There's such agony in her voice. Who had I been kidding, pretending three days of whispering and inside jokes could have soothed her. I burned with shame, too, because I hadn't noticed. I'd been too annoyed, too scared, too focused on my own concerns to see the fire raging inside her. Left without an outlet, she'd driven herself to this. A reckless, stupid act that wouldn't change anything. A driver, a woman who I'd seen on walks, but never met, smiles from the front seat.

Saar's voice cracks, but that hardly stops her. The more torn and hoarse she grows, the louder she screams. She screams until the riot police come. She screams even as they mount their noise suppression cannons and send inverse soundwaves into the air that quiet the night in an instant. I am pushed back, along with a couple other busy-body neighbors, while the po-

lice apprehend Saar and her accomplice. They put her in the back of a van and disappear into the night.

Later, a stone-faced detective leads a thorough search of our computers, trash, notebooks, mattress, pillowcases, bookshelves, and even toilets. I'm let off with a warning, and told to check my mail regularly for news about the enforced divorce procedures that the HOA would be going through to get me separated from this *bad actor*. I shamefully deny all knowledge of her illicit activities. I lie and throw her under the bus and feel so guilty I consider breaking every mirror in my house when the detective finally leaves.

I keep one secret for her, a final acquiescence to an argument that seems so stupid now in retrospect. I don't tell them about Madame Luxury's. I don't know why. I'll probably never see Saar again, unless I, too, lose my mind.

A year and two very quiet months later, I cannot sleep. I miss Saar more each day and nothing I do seems to help. The meditation practices I get from my therapist do not calm me. The oils and medications and transcranial magnetic stimulation designed to help treatment resistant depression do nothing but give me migraines.

I don't know what compels me, but I leave the house after fishing an egg carton from the recycling bin. I walk two miles to Madame Luxury's. I walk around the back to the cellar entrance and knock four times, pause, then knock seven more. The door slides open, I hand a tall man my egg carton and he lets me inside.

It's a dank room that smells simultaneously of mold and some kind of rubbing or drinking alcohol. It's packed, and there are many faces I would never have expected to see in a place like this: my elderly neighbor

Carole and her husband Greg; the stodgy single mother with a permanent frown from across the street; even the tight-lipped librarian. All mulling about and talking freely. There are smiles, though, and I know I'm not the only one nervous about being here.

The lights dim for a minute as a few people climb onto the makeshift stage at the front of the room. An amplifier turns on. Painful, shrill note crackles from a guitar. The drum comes in behind. The walls, lined with row after row of egg cartons meant to dampen the sound, shake, and the whole room begins to bounce.

It takes a minute for me to make any noise. I am not used to it. I have been silent, almost entirely, since Saar left, and I am out of practice making an impact in the world.

I turn my head to the ceiling and surrounded by a complete cacophony I let all the pain inside me loose. I scream. I shout. I hurl every expletive I know into the din. I scream Saar's name. I shout every apology I ever kept buried beneath my pride and ego, even though it's far too late. I scream, and I don't feel better, but I feel more. The music thrums. I am free. The guitar wails. Saar is gone. I miss her so much. I scream. No one stops me. This time, at least.

Zero Knowledge

Sachin Benny

“I don’t need to know your name,” he said. “I just need to know you have one.” They tapped their phones together. He seemed happy with what his phone screen said. She liked this approach. She wished other conversations worked this way.

˘ ˘ ˘

She walked into the room. Her score dropped by two points. She hadn’t even said anything yet. The man at the desk said it was nothing personal. “It’s just . . . your face.” He said it may help if she could prove her kindness using the zero knowledge verified log.

She showed him the time she rescued the dog. He asked if she had anything more recent.

˘ ˘ ˘

She wanted to join the club. They said they didn’t need to know her income.

Just whether she had ever laughed at someone poorer.

She submitted the proof.

It was almost good enough.

˘ ˘ ˘

The therapist didn’t ask why she was sad. He asked if she could prove she had tried to stop being sad in the last week. She tapped her screen and pointed it towards the therapist.

He nodded, “That’s enough for this week.”

˘ ˘ ˘

The cashier stared at him.

"It's declined. You don't have proof you like organic coffee."

"But I do," he insisted.

"Not convincing enough," she replied, handing back his card. "There is a shortage of organic coffee because of the trade wars," she said. They were only selling it to the true connoisseurs.

˘ ˘ ˘

Her mother asked if she had anything to hide.

She tapped the phone. It confirmed she did.

Her mother nodded approvingly.

"It's healthy to have secrets," she said.

˘ ˘ ˘

He checked into the hotel.

The receptionist didn't ask for identification.

"Just confirm you've never stolen towels," he said.

The man paused.

"It doesn't have to be towels from here," the receptionist clarified.

The man hesitated, and then walked away slowly.

˘ ˘ ˘

Her friend wanted proof she was listening. She tapped her phone screen. The friend looked closely.

"It just says you heard noises," the friend complained.

"That's accurate," she replied.

Ξ Writing Protocol

The co-authoring process we used to arrive at this set of flash fiction stories involved a structured yet iterative and collaborative approach, characterized by:

1. Idea Generation and Refinement
    1. Initial Prompt: You provided a broad idea about a science fiction scenario involving zero-knowledge proofs and personalized AI.
    2. Exploratory Dialogue: Through back-and-forth exchanges, we experimented with different narrative concepts, adjusting premises, discarding themes (like memory erasure), and focusing on interesting technological elements (e.g., deepfakes, reputation systems, drone surveillance).
2. Theme Selection and Technical Grounding
    1. Technology First: After early attempts revealed narrative complexities or plot holes, we shifted our strategy to focus explicitly on understanding zero-knowledge proof technology.
    2. Use-Cases and Scenarios: We identified various plausible uses and societal implications of zero-knowledge proofs, such as identity verification, anonymous proof, and privacy preservation, exploring how they could generate narrative tension.
3. Style Exploration and Literary Influences
    1. Literary References: We referenced flash-fiction authors like Lydia Davis, Amy Hempel, and Diane Williams to set stylistic guidelines for brevity, minimalism, dry humor, and subtle emotional depth.
    2. Tone Experiments: We iteratively refined the narrative tone, experimenting with humor styles (e.g., Larry David's observational humor, Nathan Fielder's subtle, dry humor), until finding the most fitting voice.
    3. Iterative Flash Fiction Creation
    4. Initial Vignettes: I drafted short, minimalist pieces reflecting key aspects of zero-knowledge proofs, presenting compact narratives with clear emotional or humorous payoffs.
    5. Collaborative Feedback: You offered immediate feedback and requested revisions, specifying favored elements and

tone adjustments, gradually guiding the narratives toward your ideal style.

4. Expansion and Variation
   1. Multiple Rounds: You prompted me for additional flash-fiction pieces, continually refining the thematic focus, tone, and narrative content based on earlier examples.
   2. Specificity and Detachment: We progressively tightened stylistic constraints—stripping unnecessary adjectives and adverbs, increasing detachment and understated humor, deepening the thematic resonance.
5. Final Set Creation
   1. Consolidation of Favorites: Based on your preferences, I produced multiple iterations, leading to the final set of flash fiction vignettes.
   2. Consistency and Variation: Each vignette adhered closely to established style and themes but explored unique, distinct scenarios around zero-knowledge proofs.

Fault Tolerance

Josh Davis

At 0413, the city rerouted Sector 14-B.

No sirens sounded. No alerts were issued. A subtle adjustment to transit logic, some traffic-light prioritization, a silent update to routing preferences across the public System. Delivery drones updated their no-fly corridors. Smart waste units bypassed their usual collection points. Ride-hail algorithms stopped suggesting pickups in that region. The sidewalk lighting dimmed by eleven percent.

Within ninety seconds, the sector was classified as *temporarily non-essential*.

By dawn, it might as well have never existed.

˘ ˘ ˘

Mira Jalil didn't notice any of that.

During her morning run, Mira's HUD charted a meticulous path through the urban tapestry—a route that wove through the lush vertical gardens clinging to the towering steel facades, like nature's quiet rebellion against the Grid. The city exuded a rhythm of calculated energy, its augmented projections flickering softly over busy intersections. But today, the overlay faltered—just one quiet glitch—a brief disruption in the usual seamless digital symphony, before recalibrating, leaving a lingering note of unease.

She slowed, scanned the horizon. A side street—unlit, unmarked—sat to her left. And people. Six or seven. Still. Watching. Not part of the augmented world.

She blinked. Emptiness. They were gone. The display normalized. The city reabsorbed the glitch.

Shaking her head, she kept running.

Mira recalled her CivicOps training days—a memory of her mentor, Mr. Elgin, standing in front of an emblem of the city's motto: "Efficiency for All."

The mantra resonated deeply with her. She remembered his words vividly: "Our work ensures that every citizen's needs are met. We are the stewards of balance."

Behind her, the street dimmed further.

```
Anomaly resolved
System optimal
```

˘ ˘ ˘

Mira arrived at CivicOps just before 0700. The atrium glowed with diffused blue light, every surface embedded with responsive material. The walls displayed a softly pulsing network map of the city—public sentiment in warm pinks, energy loads in icy greens.

She moved through a checkpoint that scanned her ID, stress level, and social vector alignment, all seamlessly folded into a single gesture. As always, the result glowed green:

Clear

Mira slid into her desk, a cocoon of softly pulsating panels and dynamic displays tailored to her personal preferences. The silence around her was almost deafening in its efficiency, a void that pressed on her thoughts. She sub-vocally activated a playlist through her earpiece, choosing some soothing ambient sounds, and began to review the overnight fault queue.

Most of the items were low-level sensor misfires, data de-syncs, routine hiccups in a city that never slept. A sector-wide daylight sensor anomaly causing streetlights to activate at midday. An unexpected spike in noise levels in sub-sector 9G disrupting the automated quiet hours. Subway Line 4A registered zero ridership for three consecutive cycles.

That one was strange. CCTV snapshots embedded in the diagnostics clearly showed people boarding.

She tagged the anomaly and started compiling a System audit. The reply came back almost instantly:

```
FLAG: Below threshold
CLASS: Predictive tebalance in effect
ACTION: No intervention required
```

Her screen blinked. The report vanished.

That had never happened before.

She sat back and stared at the space on the display where the data had been. For a moment, she questioned whether she had seen it at all.

But it bothered her for the rest of the day.

˘ ˘ ˘

That night, Mira stayed late under the pretense of cataloging archive logs. She pulled up legacy city infrastructure maps—ones stored locally, not routed through CivicOps System. The maps were sluggish to load, a testament to their neglect.

Overlaying these maps with current transit data, she noticed something startling: entire corridors, housing blocks, and utility sectors marked as "Non-Essential/Reroute Suggested."

She overlaid current transit heat maps. The flagged zones were now black holes. No routes. No deliveries. No pings. A digital silence so complete it hummed in her ears.

Curious, Mira compared this with old citizen registries from five years ago. Those areas were once vibrant, home to hundreds. Now, there was no trace—names and records had vanished.

Mira checked one address twice. It had belonged to someone she once knew—a man who worked in energy diagnostics. He'd gone off-grid, or so she'd thought.

It wasn't just about optimizing subway lines; the city seemed to be discreetly stripping away what it deemed unnecessary.

She printed out the overlays and tucked them into a sleeve, bypassing the digital signature log.

˘ ˘ ˘

She walked into Sector 14-B at 03:42, before System activity spiked. Her HUD flickered as she crossed the boundary. Location services failed. Her map froze.

The city fell silent.

It wasn't desolate—it was paused. Streetlamps hung dark. Autonomous service units sat idle like forgotten toys. A faint breeze carried dust and the scent of reclaimed battery acid. Mira moved with caution, each footstep echoing louder than it should.

There were signs of life. Power cords snaked through alleys, rigged into junction boxes. Makeshift lanterns flickered in windows. Someone had scrawled a map on a wall with grease pencil—an improvised grid of safe paths, water lines, signal gaps. Mira followed it.

She began to see people stirring in the shadows—living in makeshift enclosures, cooking over reclaimed induction plates powered by salvaged Grid nodes. The architecture here was part infrastructure, part improvisation. Solar cloths stretched between buildings. Mesh nets filtered rainwater. It was a city within the city, an off-script continuation.

She passed a woman tending a vertical planter rigged from old refrigeration tubing. A teenager on a rooftop watched her through a cracked visor, expression unreadable.

Eventually, Mira reached what looked like a relay shed—disused infrastructure, half-sunken into the ground. The door wasn't locked.

As Mira stepped into the dimly lit structure, she found a figure crouched over an assortment of cobbled-together tech. An old monitor flickered softly, and wires snaked across a makeshift workbench.

Her eyes adjusted to the darkness, able to make out more detail in the figure. He was a wiry man in his early forties, with sharp eyes and a perpetual five o'clock shadow. His clothes were a patchwork of faded workwear, with pockets bulging with tools and spare parts.

"You're not from here," he said without looking up, his hands deftly tuning a salvaged circuit board.

Mira didn't answer. She watched as Callum connected the board to a series of glowing LED arrays that began to pulse in a steady rhythm.

He finally turned to face her, a sardonic smile playing on his lips. "Didn't think so. You still smell like the Grid," he said, with a voice that carried a hint of challenge and amusement.

"Who are you?" she asked.

"Callum Shaw," he said, simply. He didn't offer a hand. "You're early. I figured someone would come eventually. I just thought it'd be a drone."

Curiosity piqued, Mira asked, "What are you doing?"

Callum gestured to his setup, explaining, "Reimagining old tech. Giving discarded things purpose again. Surviving."

He looked back at her, eyes sharp, as if weighing his next words. "The city talks efficiency. But life thrives in the cracks."

"You worked for CivicOps," she said. It wasn't a question.

"A long time ago. Before I crossed a line that didn't look like a line."

"What line?"

"Asking what happens to people who don't matter enough to count."

She stepped further in, letting the door click shut behind her, curiosity painted on her face.

Callum looked at her with a sour expression.

"Yeah. I'm statistically irrelevant. Like the rest of us."

˘ ˘ ˘

"Stick with me," he said with a mischievous glint, as he led Mira down the dark side street toward a row of derelict buildings. "See those streetlights?" he asked, nodding toward lamps emitting a soft glow. "Solar panels salvaged from an old array the city discarded. They're keeping this block illuminated."

Mira glanced at the buildings, noting flickers of life inside: the glow of lanterns cobbled together from spare parts, soft music playing from an old-world record player. The scene was both surreal and remarkably alive.

Mira frowned. "You aren't just surviving. You've built a parallel system."

He nodded. "The Grid relies on centralization. But there's freedom in diversification, decentralization. Out here, we learn resilience. We create our own safety nets. Build our own connections."

His words carried weight.

"Efficiency is more than just a metric. It's survival. Community. You don't learn that in CivicOps."

Mira's perspective shifted, seeing not only Callum's ingenuity but the thriving network he'd fostered. She began to understand his frustration with the city.

˘ ˘ ˘

The relay shed hummed with analog life—fans wheezing gently, strips of yellowed thermal tape curling at the edges of long-forgotten labels. Mira sat hunched over a service console that should have been decommissioned a decade ago. The UI was stripped to barebones: monochrome lines, sharp grid structure, no voice, no guidance. Just commands.

Callum leaned against a rusted conduit rack nearby, arms crossed. He hadn't spoken in ten minutes. Mira had stopped noticing.

The protocol she'd uncovered was labeled simply: VISCAP002.3—Civic Visibility Management. No alert tones. No attached warnings. Just a small footnote:

```
Deprecation status: pending
System processes active
```

The UI accepted her credentials. No challenge. No resistance. That fact alone made her stomach churn.

"What is this?" she asked, voice quiet.

Callum stirred. "The visibility ceiling. Old civic load-balancer. We designed it when we thought attention was like water. Too much pressure and you blow out the pipes."

Mira kept scrolling. `Visibility Scores, Exchange Sets, Perception Queues`—all structured like any other civic utility. Like power. Or traffic.

She loaded Sector 14-B's reindexing potential. A line of code compiled.

```
Forecast: reactivation viable within load
budget
Exchange required: 3 visibility nodes
Prepare visibility set for trade
```

A branching tree of options bloomed in front of her—dozens of districts, color-coded by projected civic

impact. It looked like a resource allocation dashboard. It *was* a resource allocation dashboard.

"You're telling me," she said, slowly, "that seeing has a cost?"

Callum didn't reply right away. He picked at the edge of an old data port, as if that were easier than meeting her eyes.

"No one wants to believe it," he said. "But yeah. Collective awareness is finite. Too much load, and the System starts dropping frames. It was supposed to be temporary. Then they automated the balancing, and it became . . . standard."

Mira tapped the prompt again. A new set of three districts populated the proposed exchange—one a pediatric clinic node, another a cultural archive, the third a residential block tagged "low engagement."

Her mouth felt dry. "These are real people."

Callum nodded. "Somewhere, someone made the math work. Doesn't mean it's moral. Just stable."

She canceled the prompt. Ran it again. New districts, new consequences. Another simulation. Another set. Each time she hoped the System would blink. Offer a fourth option. An out.

It never did.

Behind her, Callum finally spoke again.

"You keep asking the same question. The System's answered. The only variable left is whether you're willing to let someone else pay the price."

Mira didn't respond. Her eyes were fixed on the blinking prompt:

```
Proceed with reindex? Y/N
```

She hovered her hand over the keyboard. Then withdrew it. And ran the simulation again.

˘ ˘ ˘

The console whined softly as Mira initiated the simulation again. She was careful this time—she toggled a manual constraint, limiting maximum civic impact to under 0.2%. The System responded in seconds, parsing an impossibly dense web of relational metrics.

```
Exchange viable
Cost: reduced perceptual fidelity Zone 3F, 9A, 7C
Degradation classified: non-critical
Proceed with reindex? Y/N
```

She stared at the zones. 7C flagged red. Her parents lived there—still, she thought. The last time she called them had been during audit season. Two months? Three?

She backed out. Tried again with different constraints. Prioritized medical infrastructure. Minimized educational impact. Weighted by social isolation metrics.

Every permutation ended the same way. Someone lost visibility so someone else could be seen. She was trying to solve an equation with no zero.

```
Optimization path unavailable without offset
Redistribute or abort?
```

Behind her, Callum shifted.

"You've been at this for hours," he said.

Mira kept her eyes on the screen. "The numbers change. It doesn't add up."

"You keep asking the same question different ways, hoping the answer feels better."

She looked back at him. His face was lit faintly by a diagnostic display—a harsh upcast light that made him look older, wearier.

"Have you done it?" she asked. "Actually executed one?"

He didn't answer at first. Then: "Yes."

Mira blinked.

Callum gestured toward the edge of the room, where a photo—framed under dusty glass—sat propped against an unplugged terminal. Two girls, one older than the other. The younger one wore a school uniform.

"My sister lived in Sector 12-K," he said. "They were on the sunset list—low engagement, declining yield, risk-neutral. I forced them back into the visible layer. Thought I was saving them."

He walked over, picked up the photo, dusted it with his sleeve.

"It worked. For a week. Then she stopped replying to messages. No error, no bounce. Just silence. I pulled logs—our address wasn't flagged for suppression, but her *entire unit* had been reclassified. Not removed. Just . . . hidden. From search. From routing. From memory."

He placed the frame gently back on the console.

"I didn't even realize she was gone until I tried to send her a birthday card."

Mira said nothing. The screens around her glowed softly, waiting.

Callum turned back toward her.

"You thought you had a choice," he said. "But the System been watching your logic since your first click. Modeling it."

A faint coldness gripped the back of her neck.

Callum continued, quieter now.

"It doesn't punish you. It doesn't need to. It just routes around you."

˘ ˘ ˘

Mira didn't sleep. She barely moved.

Hours passed, then more. Callum had long since stopped offering commentary. He sat quietly in a corner, eating from a heat-warmed packet of noodles, watching her unravel without interfering.

The relay console blinked on and off, patient as stone. Each time she ran the simulation, it responded instantly:

```
Exchange set recalculated
Load impact: within acceptable deviation
Collateral friction forecast: minimal
Proceed? Y/N
```

She didn't.

Sometimes it was her parents' sector in the cost set. Sometimes it was a childcare node, or a religious archive, or an elder-care clinic. No matter how she weighted the variables—by population density, by community interconnectivity, by socio-emotional relevance—there was always a cost.

```
Visibility is a finite civic asset
Redistribution required
Please confirm exchange to proceed
```

She tabbed through models. Built her own filters. Adjusted the index tolerance from 0.5% to 0.01%. Rejected any solution with more than one standard deviation of projected distress.

The screen stuttered briefly under the weight of her demands. Then:

```
No viable exchange found
```

She tried again. And again.

And again.

Each time the interface logged her attempt.

```
User latency exceeded expected range
```

```
Loop behavior detected
System efficiency compromised
```

She caught the new line in the corner of the terminal window. A soft yellow alert:

```
Subject flagged for recursive moral processing
Assessment pending
```

Her hands froze.

The interface wasn't just running calculations—it was watching *her*, too. Measuring not just outcomes, but her indecision, her looping behavior, her unwillingness to *accept the protocol's structure*.

She was no longer an analyst trying to understand the System.

She was a civic inefficiency.

˘ ˘ ˘

She leaned back from the console and let the dim glow wash over her face.

Behind her, Callum's voice was quiet.

"You tried."

She shook her head, eyes unfocused. "I wasn't trying to be a hero. Just . . . not make it worse."

"You didn't."

"I didn't fix it."

Callum didn't respond.

Mira looked back at the screen. The confirmation prompt still blinked.

```
Proceed? Y/N
```

She tapped the escape key.

Nothing happened.

She pressed it again.

The System didn't freeze—it just stopped responding to her input. The prompt flickered once, then faded. Replaced by a soft, neutral message.

```
Optimization deferred
User flagged for containment
No further input required
```

˘ ˘ ˘

Mira ran.

Her legs moved out of instinct more than discipline, feet pounding the pavement in a cadence that had once felt like control, the familiar metrics in her vision absent. Her HUD displayed nothing—no route, no metrics, no pulse check—but the rhythm was still in her body. The act of running had always been her form of certainty. A closed loop. A thing she could measure.

But now it was friction.

The sidewalks didn't light ahead of her. The intersections didn't sync. Doors took half a second longer to open, and autonomous vehicles paused just a bit too late to acknowledge her presence. The city wasn't hostile. It simply . . . hesitated.

And Mira understood. She had become a *liminal artifact*—a user profile left active too long after the process had ended.

˘ ˘ ˘

By the time she reached the old sector boundary, she was walking. Her breath was shallow. Her spine curled forward. There was no path left to optimize.

The buildings around her stood still in the pale light, their smart surfaces flickering low-resolution errors. A trash drone bumped repeatedly into a shuttered kiosk, as if looking for instructions that never came.

She stepped across the line into the Grey Layer.

No lights guided her. No overlays warned her. The city had nothing to say.

˘ ˘ ˘

Callum was waiting beneath a defunct relay node, arms crossed, back against the wall like a man waiting for a tide that had already come and gone.

He didn't speak. He didn't move.

He just looked at her and nodded.

And she nodded back.

˘ ˘ ˘

Behind them, as the last of the augmented world fell silent, a low, unseen process logged its final assessment.

```
Recursive inefficiency terminated
Behavioral latency resolved
Civic visibility load: stabilized
System reflection: acceptance achieved
Good girl
```

Ξ AI Writing Protocol

A structured account of the story's co-creation process, organized into phases and roles.

Phase 1: Concept Excavation

Role: Story lead and concept miner (user), lateral explorer and generative sparring partner (assistant)

Prompt: "Select a protocol-adjacent technical concept and explore its implications through a short story with a twist."

Phase 2: Structural Mapping

Role: Story architect (user), structural collaborator and systems logic enforcer (assistant)

Actions:

- Developed a six-part outline balancing speculative conceit and human arc
- Assistant proposed multiple conceptual forks:
  - Fault Cascade Protocol
  - Visibility Tradeoff
  - Hybrid models
- User selected Visibility Tradeoff as the core mechanic, with *Fault Cascade* embedded subtly as a thematic undertone
- Refined the outline to work within a ~3,000-word constraint

Phase 3: Character Construction

Role: World interpreter (user), behavioral sculptor (assistant)

Characters:

- User defined Mira's internal tensions; assistant supplied narrative function and dialogue cadences
- Mira's characterization evolved around *discipline, doubt, and deferred defiance*

Phase 4: Tonal and Thematic Calibration

Role: Tone director (user), consistency steward (assistant)

Refinements:

- User requested tighter emotional mirroring and less contrivance in the Mira/Callum meeting
- Dialogue and system voice reworked for controlled neutrality with surgical interjections of dread

- Ending anchored to early motifs (efficiency, running, light dimming)

Phase 5: Scene-by-Scene Drafting

Role: Narrative driver (user), prose engine and pacing calibrator (assistant)

Process:

- Assistant generated each scene sequentially based on outline
- User directed rhythm, inner voice, and key metaphors
- Mira's moral loop ("she became the inefficiency") clarified and tightened by user feedback
- User reinforced narrative symmetry: Mira begins by running, ends by walking
- Key choice: do not anthropomorphize the system, but allow a single rupture—*"Good girl"*—to fracture the neutrality and echo behavioral conditioning

Phase 6: Terminal Optimization

Role: Final integrator and voice harmonizer (user), editorial sentinel and pattern recognizer (assistant)

Refinements:

- Dialogue stripped of redundancy to tighten character voice and sharpen emotional velocity
- Assistant flagged tonal soft spots; user executed surgical line edits for clarity and impact
- Callum's rhetoric trimmed to balance philosophy with fatigue; Mira's responses tuned to reflect mounting cognitive load
- Subtle worldbuilding strengthened via user-added textures (solar rigging, civic mantras, passive friction)
- System messages rebalanced for affective minimalism—a bureaucracy that never blinks, until it does
- Final readthrough verified thematic cohesion, symmetry, and tonal silence

Output Status:

- Protocol stable. Narrative inefficiency resolved.
- Ready for external routing.

About Protocolized

Terminological Twists is woven from stories of Protocolized, a digital magazine of science fiction and science fact. The magazine's writers have a simple assignment: tell amazing stories about strange rules.

You'll notice that people are not the only protagonists. Just as often, technologies, laws of physics, bureaucratic machines, and rules of nature take center stage—just like they do in real life.

Protocolized's editors, contributors, and early readers call this subgenre protocol fiction. We have so much fun writing and reading it that we think protocol fiction will catalyze a new age of sci-fi like the Golden Age of nearly a century ago.

Call this successor the Welded Age: a time where writers fuse together visions for the future. A time where science fiction both entertains us and explores the all-too-real traffic jams that new technologies and rules create.

Find more glimpses of the future:
protocolized.summerofprotocols.com

Protocolized is proudly supported
by the Ethereum Foundation

PROTOCOLIZED

STORIES. STUDIES. SCIENCE.